

ZIYAO SU

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EDUCATION

The Ohio State University	Columbus, OH
Ph.D. in Materials Science and Engineering GPA: 3.96/4.00	2023 – 2027
M.S. in Materials Science and Engineering GPA: 3.94/4.00	2023 – 2024
B.S. in Physics and Mathematics, Minor in Music GPA: 3.73/4.00	2018 – 2022

TECHNICAL SKILLS

- ★ **Research Expertise:** Ultrafast Laser–Matter Interactions; Relativistic Laser–Plasma Interactions; Laser-Induced Damage; Multiphysics Electromagnetic Modeling; Inverse Design of Optical Components.
- **Computational Methods:** FDTD; PIC (EPOCH); RCWA (RETICOLO); DFT (VASP); Bayesian Optimization.
- **High-Performance Computational Platforms:** Unity Cluster; Ohio Supercomputer Center (OSC); NERSC.
- **Programming Languages:** Python; MATLAB; C++; Java.
- **Software Tools:** Mathematica; VisIt; CrystalMaker; SNLO.

RESEARCH EXPERIENCE

Graduate Research Associate, Femto-Solid Laboratory, OSU	Sep 2021 – Present
★ Research Focus: Multiphysics FDTD and PIC modeling of ultrafast laser–matter interactions, laser-induced damage threshold prediction, and inverse design of optical components.	
• Independently developed 2D FDTD (3D In progress) codes in MATLAB to model femtosecond laser interactions and damage mechanisms in metallic/dielectric multilayer mirrors, diffraction gratings, and nanolaminate thin films. Implemented a Total-Field/Scattered-Field (TFSF) source and integrated multiple physical models, including Keldysh ionization, impact ionization, Drude response, Ohmic heating, and electron-collision processes.	
• Developed a hybrid FDTD–RCWA inverse-design framework coupled with Bayesian machine learning for multi-objective optimization of high-performance diffraction gratings.	
• Conducted PIC simulations with EPOCH/Q-EPOCH to investigate nonlinear electron dynamics and energy spectra in petawatt laser interactions with multiphase targets (water film; C ₂ H ₆ O ₂ film; and Ta foils); developed a Keldysh ionization model in EPOCH for femtosecond laser–solid multilayer interactions and damage-threshold prediction, and performed large-scale simulations on HPC platforms including Unity Cluster, OSC, and NERSC.	
• Collaborated with Colorado State University and XUV Lasers Inc. to experimentally validate laser-damage-threshold models for optical components including mirrors and gratings.	

HONORS

- **MJ Soileau Best Student Paper Award** (SPIE Laser-Induced Damage in Optical Materials 2025).
- **The Ohio State University Undergraduate Dean’s List** (6 semesters: Fall 2018, Fall 2019, Spring 2020, Spring 2021, Fall 2021, Spring 2022).

ACADEMIC SERVICE

- **Journal Reviewer (9 manuscripts reviewed):** *Applied Optics* (2024, 2025); *Optics Express* (2026); *Optics Letters* (2026); *Optical Materials Express* (2026).
- **Instructional Assistant:** OSU MSE 2010 – Introduction to Engineering Materials (Fall 2024); OSU MSE 3331 – Materials Science and Engineering Laboratory I (Fall 2025).
- **Professional Memberships:** American Physical Society (APS); SPIE; SPIE OSU Student Chapter.

SELECTED PUBLICATIONS

Journal Articles

- **Ultrafast Laser Interaction with Dielectric–Metal Hybrid Coatings: A 2D FDTD Modeling Study**
Ziyao Su, and Enam A. Chowdhury. *High Power Laser Science and Engineering* **14**, e78 (2026).

- **Pulse-Duration Scaling of Laser-Induced Damage Threshold in Hybrid Gratings under Ultrafast Irradiation.**
Ziyao Su, and Enam A. Chowdhury. *Optics Letters* **51**, 4236–4239 (2026).
- **A Fully Three-Dimensional Kinetic Particle-In-Cell Framework for Modeling Laser-Dielectric Interactions: Few-Cycle Pulse Damage.**
Joseph R. Smith, Ziyao Su, Simin Zhang, Charles Varin, Vitaly E. Gruzdev, and Enam A. Chowdhury. *High Power Laser Science and Engineering* **13**, e60 (2025)
- **Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings.**
Simin Zhang, Ziyao Su, Carmen S. Menoni, and Enam A. Chowdhury. *Optics Letters*, **48**, 1212–1215 (2023).

Conference Proceedings

- **FDTD Modeling of Femtosecond Laser–Hybrid Grating Interaction and Damage Mechanisms.**
Ziyao Su, and Enam A. Chowdhury. *CLEO 2026, Technical Digest Series, paper JW1.123* [Poster].
- **Keldysh ionization based-FDTD modeling of laser-induced damage threshold of MLD-IBS compression gratings for petawatt 2 μm laser systems.**
Ziyao Su, Mohamed Yaseen Noor, Emma DeAngelis, Carmen S. Menoni, Enam A. Chowdhury. *Proc. SPIE PC13729, Laser-Induced Damage in Optical Materials 2025, PC1372907 (19 December 2025)* [Talk].
- **Electron dynamics of femtosecond laser induced plasma inside multilayer dielectric high reflectors studied using particle-in-cell (PIC) approach.**
Ziyao Su, Joseph R. Smith, and Enam A. Chowdhury. *CLEO 2024, Technical Digest Series, paper SF3H.5* [Talk].

PRESENTATIONS

- **FDTD Modeling of Sub-Picosecond Laser Interaction and Damage in Hybrid Optical Coatings.**
Ziyao Su, and Enam A. Chowdhury. *Proc. SPIE 14074-47, High-Power Laser Ablation IX, 14074-47X (March 2026)* [Poster].
- **2D FDTD Modeling of Ultrashort Laser–Matter Interactions.**
Ziyao Su, and Enam A. Chowdhury. *Laboratory for Laser Energetics (LLE), University of Rochester, Nov. 5, 2025* [Invited Talk].
- **Electron Dynamics of Femtosecond Laser-Induced Plasma Inside Multilayer Dielectric High Reflectors Studied Using a Particle-in-Cell (PIC) Approach.**
Ziyao Su, Joseph R. Smith, and Enam A. Chowdhury. *LaserNetUS User Meeting, 2024* [Poster].

EXTRACURRICULAR ACTIVITIES

Athletics	Jun 2025 – Present
<i>OSU Table Tennis Team / Columbus Table Tennis Club</i>	
• Varsity team member and registered USATT player (USATT Rating: 1956); competed in NCTTA tournaments.	
Student Organizations	Sep 2024 – Present
<i>Chinese Students and Scholars Society (CSSS) / Puzzleology Club</i>	
• Event planning, promotion, and organization for activities with dozens to over one hundred participants.	
Campus Employment	Nov 2018 – Dec 2019
<i>Traditions at Kennedy / The Blackwell Inn and Pfahl Conference Center</i>	
• Student cook and assisted with the preparation and cooking of vegetables, meats, eggs, pancakes, and other foods; operated dishwashing systems and maintained kitchen sanitation.	